



BACHELORARBEIT

ON CAUSAL DISCOVERY FOR DEEP LEARNING IN A BIOLOGICAL CONTEXT

Verfasser
Trayan Tsonev

angestrebter akademischer Grad
Bachelor of Science (BSc)

Wien, 2024

Studienkennzahl lt. Studienblatt: A 033 521

Fachrichtung: Informatik

Betreuer: Assoz. Prof. Dipl.-Ing. Dr.techn.
Sebastian Tschatschek, BSc

Abstract

We explore different causal discovery algorithms on the hidden states and outputs of deep learning models trained to predict high/low protein-to-RNA ratio (PTRs) in different tissues based on cDNA sequences. The dataset we used contained cDNA sequences with heterogeneous properties, such as varying lengths and inclusion of coding (CDS) and untranslated (UTR) regions. The target values were for each sequence based on precalculated protein-to-mRNA continuous values. Two primary model categories were investigated: RNN-based and CNN-based models. The RNN-based model, consisting of 4 GRUs followed by dense layers, showed promising results regarding prediction of the PTRs, whereas the CNN-based models did not perform beyond chance level. Consequently, only RNN outputs were further analyzed using causal discovery algorithms. These algorithms, including LiNGAM, FCI and GES, were applied to identify causal relationships between different segments of the RNN output data, the tissue type, the model prediction and other variables. ICA-LiNGAM showed the most consistent results based on ad-hoc comparisons, identifying the UTRs and segments from last parts of the CDS as having the most influence on the model prediction and other variables.

Contents

1	Introduction	4
2	Related Work	4
3	Methodology and Experimental setup	5
3.1	Dataset	5
3.2	Models	5
3.2.1	RNN-based model	6
3.2.2	CNN-based model	8
3.3	Causal Discovery Algorithms	8
3.3.1	LiNGAM	8
3.3.2	FCI	8
3.3.3	GES	9
3.3.4	Additional Algorithms	9
4	Evaluation and Discussion	9
4.1	Exploratory data analysis	9
4.2	Amino acid frequencies	10
4.3	Coding sequences	10
4.4	Coding sequences and untranslated regions	11
5	Conclusion and Future Work	14
5.1	Graph similarity assessment	14
5.2	Autoencoders	14
5.3	Transformer Architecture	14
5.4	Alternative Causal Discovery Techniques	14

1 Introduction

In recent years, advances in the field of AI (and more specifically deep machine learning) have enjoyed significant attention by researchers, industry and the wider public [11] [8]. However, most of these models can be considered as "black boxes", leaving little to no room for interpretation, i.e. being hard to comprehend due to the large number of parameters and complicated dependencies [22]. This fact has limited the usefulness of deep learning in many research scenarios, where understanding the relationship between input and output is crucial. The emergence of explainable AI (xAI) and the application of various causal inference/discovery techniques has the potential to address this limitation [22].

While not as widely publicized, the field of molecular biology has also undergone significant developments over the last years [13]. After the Human Genome Project was completed in 2006 [10], personal genome sequencing has not only become possible, but also progressively cheaper and more accessible [1]. This has led to the emergence of practices such as personalized medicine and genetic disease risk assessment. Additionally, RNA therapeutics, a new class of medications, have generated promising results. All of these developments, however, often require understanding various different properties of the given RNA sequences [4].

In this thesis, we try to bridge deep learning and causality within the context of the properties of full-length RNA sequences. We train deep learning models to predict tissue expression levels and use causal discovery algorithms to identify important aspects of the input RNA sequences, a step which is not at all trivial due to the large and vastly differing lengths of the input.

2 Related Work

Previous work [7] has demonstrated the use of traditional statistical learning methods (random forests) for predicting protein expression levels in different tissues based on the encoding mRNA sequence. These ratios have been used for generating optimized versions of the codons for each of these tissues, which has shown promising results in cell-line models for therapeutic applications. Additionally, deep learning architectures (namely convolutional and recurrent neural networks [21] and transformer encoders [6]) have been shown to be effective at predicting mRNA degradation dynamics and at providing some interpretability; however, only by taking the 3'-UTR of the sequences into account. Causal discovery algorithms have been applied to molecular phenotype [9] and single-cell RNA-sequencing (scRNA-seq) data [19] which has established the feasibility of using such techniques in the context of genetics but also identified several challenges and limitations (e.g. poor scaling, ambiguity due to generated Markov equivalences, etc.)

3 Methodology and Experimental setup

3.1 Dataset

The dataset [20] consisted of 89411 cDNA sequences. Notably, The properties of the individual sequences were very heterogeneous: lengths ranged from 27 all the way up to 109224 nucleotide. Some sequences contained only the coding sequence part (CDS), while others included the 5' untranslated region, the 3' untranslated region or both. For the purposes of training the models, 29 target values (both binary UP/DOWN and continuous values) were calculated for each sequence based on the protein-to-mRNA ratios for different tissues [7].

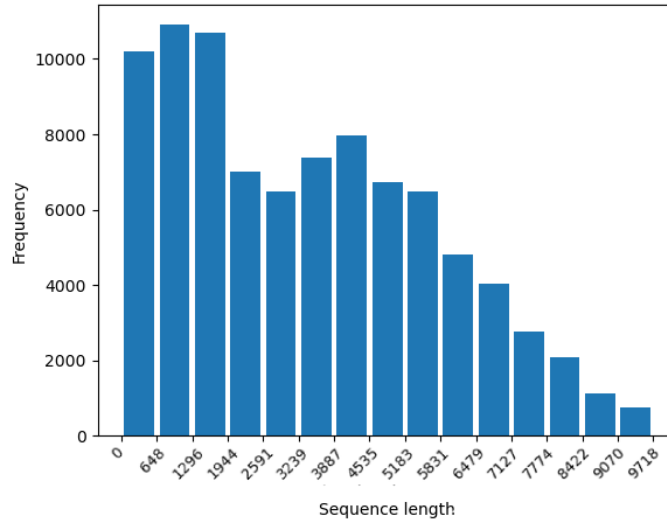


Figure 1: Histogram of the cDNA sequence lengths in the dataset

total	only CDS	CDS and 5UTR	CDS and 3UTR	CDS, 5UTR, 3UTR
89411	3759	17910	7192	60550
1.00	0.04	0.20	0.08	0.68

Table 1: Absolute and relative frequencies of cDNA sequences based on included regions

3.2 Models

Different models were trained to classify cDNA sequences as resulting in either a high-PTR or a low-PTR in 29 different tissues. The input features were one-hot-encoded/embedded cDNA sequences and the meta data (i.e. index that identifies the tissue type). Due to their differing lengths, the sequences were

padded with zeros; however the zeros (respectively the parts of the hidden states corresponding to the zeros) were discarded before being fed into the dense layers which were followed by a Sigmoid function that makes binary predictions. Many cDNA sequences were included in the training data multiple times (albeit with a different label) due to being highly/lowly expressed in more than one tissue, while others were not used at all due to not being highly or lowly expressed in any tissue. Binary cross entropy was chosen as the loss function.

For both model categories (RNN and CNN), we tested several different configurations (e.g. different number of layers / units in the hidden, kernel sizes, shapes for the dense layers, etc.) but did not observe any major differences in performance.

3.2.1 RNN-based model

For the RNN-based model category, we settled on an architecture consisting of 4 unidirectional GRUs with 32 units each. The output of the RNN was fed both into a logger function and into the dense part of the model.

The logger functions splits the RNN output into segments of equal lengths, then calculates the means of each segment and saves them.

The dense part consists of two fully-connected layers which use the ReLU activation function; a sigmoid function is used to make a binary prediction between 0 and 1.

The output of the sigmoid function as well as the means saved by the logger are then used as observational data in for the causal learning algorithms.

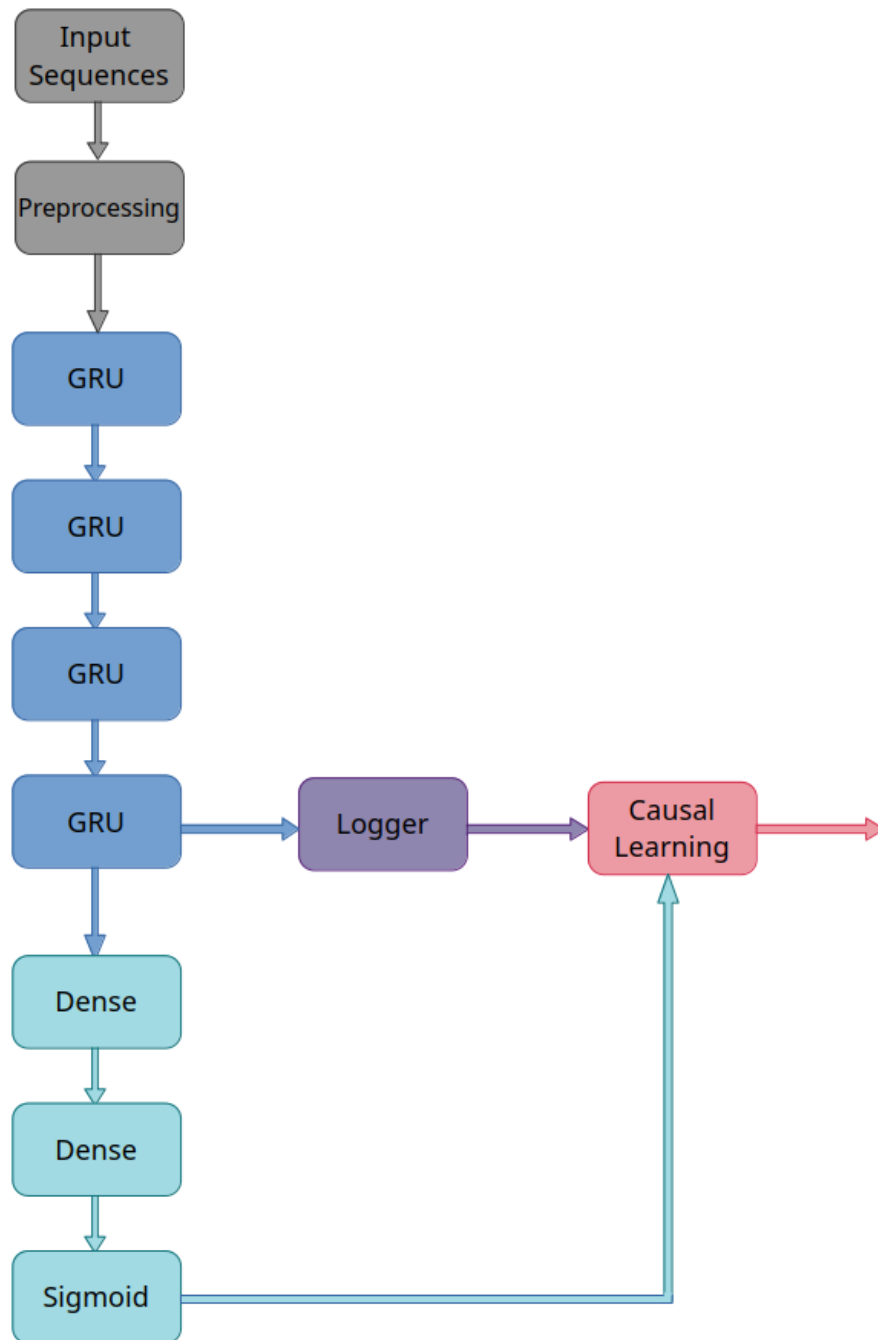


Figure 2: High-level summary of the major steps of the workflow for the RNN-based model

3.2.2 CNN-based model

For the CNN-based versions, we tried several different configurations: both with a single and multiple convolutional layers and with different stride and kernel sizes. However, performance was unsatisfactory and barely exceeded chance level. Therefore, we decided to not include the output in the causal learning stage.

kernel size	stride sizes
3	1,2,3
6	1,2,3,6
20	2,3,6,10,20
60	3,6,10,30,60
300	3, 10, 60, 100, 300

Table 2: Attempted kernel and stride size combinations for the CNN model

3.3 Causal Discovery Algorithms

Causal discovery algorithms aim to find causal relationships between variables. Unlike more traditional causal inference techniques, which estimate the effect of one variable on another [12], causal discovery tries to learn a Bayesian network representation of the relationships in the observational data and visualize it via directed acyclic graphs (DAGs) [12] or fuzzy cognitive maps (FCMs) [14]. This is usually done by assigning scores or enforcing constraints on graphs within the search space or by adding a noise term to different variables and tracking how that affects others. We attempted to identify causal relationship using a range of popular causal discovery algorithms (in order to cover different underlying techniques), namely:

3.3.1 LiNGAM

LiNGAM [15] is based on the assumptions of linearity of the data, non-Gaussianity of the noise, acyclicity and no latent common causes. We utilized both the standard ICA (Independent Component Analysis) variety as well as DirectLiNGAM [16], which converges more easily and doesn't require algorithmic parameters.

3.3.2 FCI

Fast Causal Inference [17] is a constraint-based algorithm which works by first generating a fully connected undirected graph of the data, testing all variable pairs for conditional independence (removing edges where it is detected) and finally directing the remaining edges based on detected structures.

3.3.3 GES

Greedy Equivalence Search [2] is a score-based algorithm that searches the graph space (starting from an empty graph) by assigning a Bayesian score to each graph it generates and using it as a heuristic.

3.3.4 Additional Algorithms

We attempted to also use the Peter-Clark (PC) [18] algorithm and Causal Generative Neural Networks (CGNN) [5] ; however running PC often took several hours even on a GPU and CGNN only converged when provided with a skeleton.

4 Evaluation and Discussion

4.1 Exploratory data analysis

Before training the models, we first performed exploratory data analysis on the continuous target values, in the hopes that a low-dimensional visualization of the input data might reveal some sort of an underlying relationship/structure in the data;. We attempted two techniques: PCA and t-SNE, reducing the data to 2 and 3 dimensions (missing/undefined values were filled in with zeros). PCA revealed no useful information; however, t-SNE showed a distinct cluster on the right side of the plot, indicating an internal structure within the data.

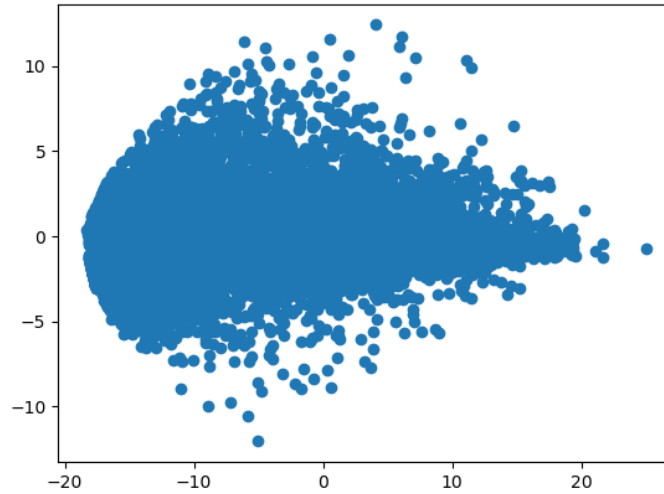


Figure 3: PCA: no obvious clustering/structures of interest can be seen

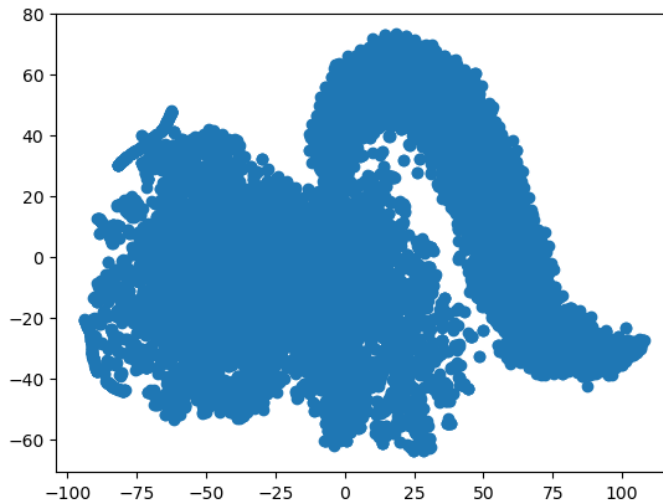


Figure 4: t-SNE: two visible clusters suggesting an internal structure of some sort

4.2 Amino acid frequencies

Before turning to the predictive models, we attempted to discover causal relationships by analysing the dataset, more specifically by calculating the frequencies of each of the 20 amino acids [synthesized by the human genome] in each cDNA sequence and using those as input data for the causal discovery algorithms, along with the target value (either high- or low-PTR). The causal structures¹ found by the different algorithms shared little to no similarities (based on an ad-hoc comparison of the generated causal graphs), other than the fact that the target value was not in the causal order (i.e. its node wasn't connected to any of the others).

4.3 Coding sequences

The bulk of the experiments were focused on trying to discover a causal relationship or different degrees of importance for the model prediction between different sub-regions of the coding sequence (CDS). We decided to focus on the CDSs at first, since they're present in all sequences in the dataset and are the part that gets translated into mRNA. To that end, the one-hot-encoded/embedded CDSs were processed through the RNN, the output of which was divided into either 4 or 8 segments of equal lengths. Then, the means of each segment were passed on to the causal learning algorithms along with the segment length and either the predicted or ground truth value. Overall, there was little inter- and intra-algorithmic consistency. Direct LiNGAM and GES often produced

¹Causal graphs have been omitted due to their size (20+ nodes).

starkly different causal graphs even on model outputs with identical parameters and nearly identical performance. FCI mostly detected common latent cause relationships. However, ICA LiNGAM was an exception and produced mostly consistent results. It assigned segments from the second half of the CDS and the predicted target value to the top of the causal order and the tissue metadata, ground truth target value and segment length to the bottom.

4.4 Coding sequences and untranslated regions

Another configuration we attempted was to pass the 5' and 3' UTRs as well as the CDS separately through the RNN, calculate the mean values for each of these regions and attempt causal discovery based on them. The results were noticeably more consistent compared to the other approaches (both between different algorithms and different runs). The UTRs were consistently identified as being on top of the causal order and very important for inferring the prediction / ground truth value, while the CDS was ranked either towards the bottom/middle or in the case of LiNGAM had negligible edge weights.

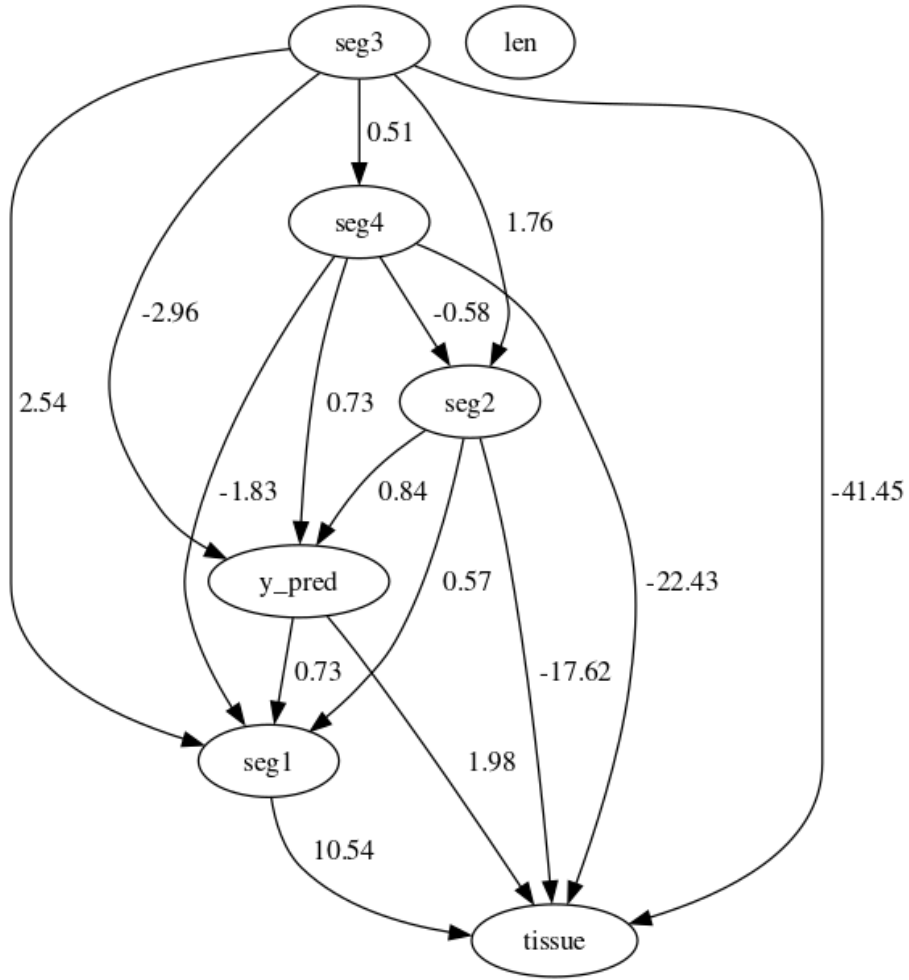


Figure 5: Typical DAG output of ICA-LiNGAM: the nodes representing the 3rd and 4th segments of the RNN output are on the top of the graph, suggesting a higher degree of importance and influence over the other variables

The numbers on the edges represent the term of the equation estimated to represent the relationship between the two nodes, i.e. a high absolute value represents a stronger correlation

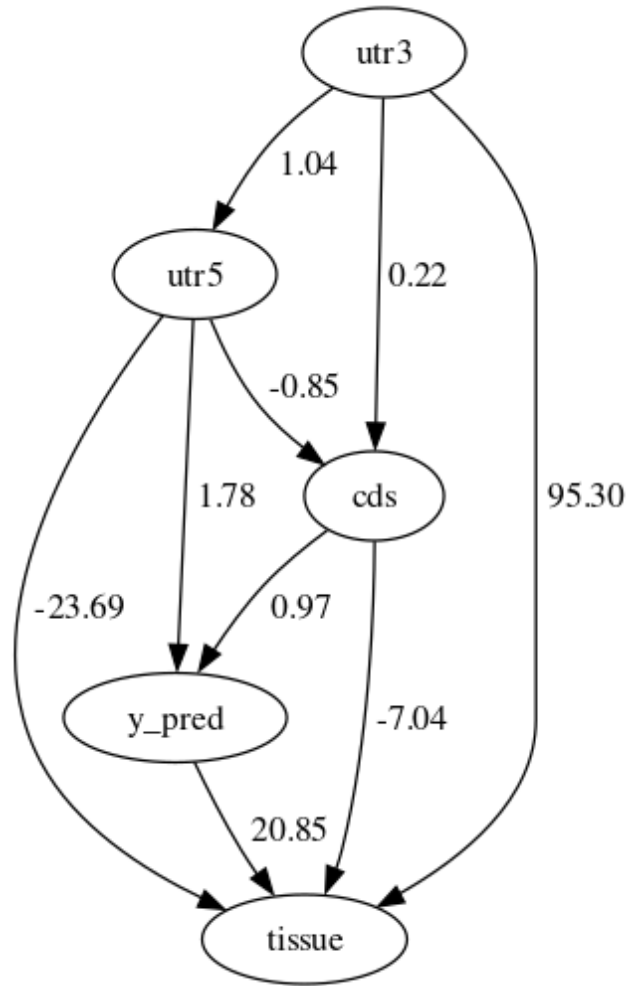


Figure 6: Typical DAG output of ICA LiNGAM with the UTRs included: the nodes representing the UTRs are ontop, suggesting a higher degree of importance and influence over the other variables

The numbers on the edges represent the term of the equation estimated to represent the relationship between the two nodes, i.e. a high absolute value represents a stronger correlation

5 Conclusion and Future Work

We trained an RNN-based model to predict cDNA sequences’ binary low/high PTRs in 29 different tissues. We ran different causal algorithms on the sequences’ amino acid frequencies and on the RNN’s output. We identified the ICA version of LiNGAM as the most consistent algorithm of all that were tested based on ad-hoc comparisons and suggest that the UTRs and the second half of the input sequences are of great importance for the model’s prediction. In future work, we intend to explore the following approaches:

5.1 Graph similarity assessment

Instead of relying on ad-hoc comparisons of the causal graphs, we suggest adopting a quantitative graph similarity assessment approach (e.g. Structural Hamming Distance or Graph Edit Distance) which would allow for a more robust and precise analysis and a more concrete way to estimate algorithm consistency.

5.2 Autoencoders

Autoencoders are a type of neural network architecture that first embed their input into a lower dimension and then aim to reproduce it as its output. We suggest that the lower-dimensional embedding might hold information that is relevant for causal learning and biological knowledge discovery. Traditionally, autoencoders consist of fully-connected dense layers, which is not a suitable architecture for inputs of starkly different sizes; however, LSTM-based autoencoders [3] might prove useful.

5.3 Transformer Architecture

Transformers are a type of neural network architecture that is based on a multi-head attention mechanism and are the main approach used in recent AI breakthrough such as ChatGPT, BERT, etc. They convert their input into a numerical representation which is then iteratively contextualized. We suggest that the architecture might be able to extract useful information from the data that can then be used as a basis for causal learning.

5.4 Alternative Causal Discovery Techniques

All algorithms that we explored within the scope of our work search for causal structures only within a discrete graph space. However, methods that work in a continuous search space [23] also exist. Building upon those approaches, deep causal learning frameworks that integrate interventional data and domain knowledge have also been developed [23]. Such methods could lead to the discovery of more consistent patterns.

References

- [1] AKINTUNDE, O., TUCKER, T., AND CARABETTA, V. J. The evolution of next-generation sequencing technologies. *ArXiv* (2023).
- [2] CHICKERING, D. M. Optimal structure identification with greedy search. *J. Mach. Learn. Res.* 3, null (mar 2003), 507–554.
- [3] DHANUKA, R., TRIPATHI, A., AND SINGH, J. P. A semi-supervised autoencoder-based approach for protein function prediction, 2022.
- [4] FORERO, D. A., AND CHAND, V. Methods in molecular biology and genetics: looking to the future. *BMC Research Notes* 16, 1 (March 2 2023), 26.
- [5] GOUDET, O., KALAINATHAN, D., CAILLOU, P., GUYON, I., LOPEZ-PAZ, D., AND SEBAG, M. Causal generative neural networks. *ICRL* (2018).
- [6] HE, S., GAO, B., SABNIS, R., AND SUN, Q. RNAdegformer: accurate prediction of mRNA degradation at nucleotide resolution with deep learning. *Briefings in Bioinformatics* 24, 1 (01 2023), bbac581.
- [7] HERNANDEZ-ALIAS, X., BENISTY, H., RADUSKY, L., SERRANO, L., AND SCHAEFER, M. Using protein-per-mrna differences among human tissues in codon optimization. *Genome Biology* 24 (02 2023).
- [8] KANBACH, D. K., HEIDUK, L., BLUEHER, G., SCHREITER, M., AND LAHMANN, A. The genai is out of the bottle: generative artificial intelligence from a business model innovation perspective. *Review of Managerial Science* 18, 4 (2024), 1189–1220.
- [9] KELLY, J., BERZUINI, C., KEAVNEY, B., TOMASZEWSKI, M., AND GUO, H. A review of causal discovery methods for molecular network analysis. *Mol Genet Genomic Med* 10, 10 (Oct 2022), e2055.
- [10] LANDER, E., ET AL. Initial sequencing and analysis of the human genome. *Nature* 409, 6822 (2001), 860–921.
- [11] MALHOTRA, R., AND SINGH, P. Recent advances in deep learning models: a systematic literature review. *Multimedia Tools and Applications* 82, 29 (2023), 44977–45060.
- [12] NOGUEIRA, A. R., PUGNANA, A., RUGGIERI, S., PEDRESCHI, D., AND GAMA, J. Methods and tools for causal discovery and causal inference. *WIREs Data Mining Knowl. Discov.* 12, 2 (2022).
- [13] ROALIN, A. Overview on recent advances in molecular biology techniques. *Adv Tech Biol Med* 10 (2022), 355. Received: 24-Mar-2022, Manuscript No. ATBM-22-17381; Editor assigned: 29-Mar-2022, Pre QC No. ATBM-22-17381 (PQ); Reviewed: 14-Apr-2022, QC No. ATBM-22-17381; Revised:

21-Apr-2022, Manuscript No. ATBM-22-17381 (R); Published: 28-Apr-2022.

- [14] SHARMA, A., TSELYKH, A., PODOPLELOVA, E., AND TSELYKH, A. Knowledge-oriented methodologies for causal inference relations using fuzzy cognitive maps: A systematic review. *Computers Industrial Engineering* 171 (2022), 108500.
- [15] SHIMIZU, S., HOYER, P. O., HYVÄRINEN, A., AND KERMINEN, A. A linear non-gaussian acyclic model for causal discovery. *J. Mach. Learn. Res.* 7 (dec 2006), 2003–2030.
- [16] SHIMIZU, S., INAZUMI, T., SOGAWA, Y., HYVÄRINEN, A., KAWAHARA, Y., WASHIO, T., HOYER, P. O., AND BOLLEN, K. Directlingam: A direct method for learning a linear non-gaussian structural equation model. *J. Mach. Learn. Res.* 12, null (jul 2011), 1225–1248.
- [17] SPIRITES, P. An anytime algorithm for causal inference. In *Proceedings of the Eighth International Workshop on Artificial Intelligence and Statistics* (04–07 Jan 2001), T. S. Richardson and T. S. Jaakkola, Eds., vol. R3 of *Proceedings of Machine Learning Research*, PMLR, pp. 278–285. Reissued by PMLR on 31 March 2021.
- [18] SPIRITES, P., GLYMOUR, C., AND SCHEINES, R. *Causation, Prediction, and Search*, 1 ed., vol. 81 of *Lecture Notes in Statistics*. Springer New York, NY, 1993. Springer-Verlag New York, Inc. 1993.
- [19] WEN, Y., HUANG, J., GUO, S., ELYAHU, Y., MONSONEGO, A., ZHANG, H., DING, Y., AND ZHU, H. Applying causal discovery to single-cell analyses using causalcell. *eLife* 12 (may 2023), e81464.
- [20] WOLFINGER, M. T. Gencode43.
- [21] YAISH, O., AND ORENSTEIN, Y. Computational modeling of mRNA degradation dynamics using deep neural networks. *Bioinformatics* 38, 4 (11 2021), 1087–1101.
- [22] YANG, W., WEI, Y., WEI, H., CHEN, Y., HUANG, G., LI, X., LI, R., YAO, N., WANG, X., GU, X., AMIN, M. B., AND KANG, B. Survey on explainable ai: From approaches, limitations and applications aspects. *Human-Centric Intelligent Systems* 3, 3 (2023), 161–188.
- [23] ZHENG, X., ARAGAM, B., RAVIKUMAR, P., AND XING, E. P. Dags with no tears: continuous optimization for structure learning. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems* (Red Hook, NY, USA, 2018), NIPS’18, Curran Associates Inc., p. 9492–9503.